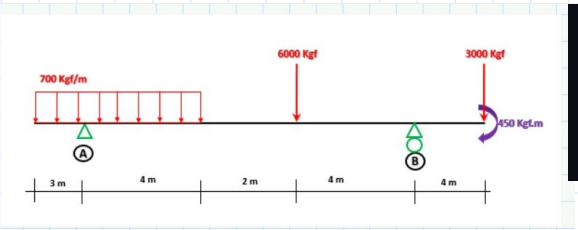
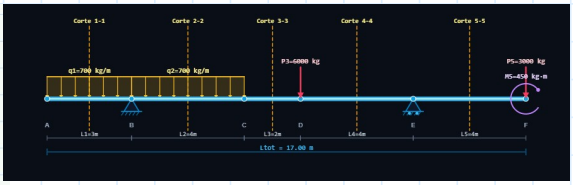
EJERCICIO 01_ ULIANOV SABATH CUBA VALENCIA

01. DIAGRAMA DE CUERPO LIBRE DCL





02. CALCULO DE REACCIONES

$\sum F_y = 0 \qquad \sum M_a = 0$

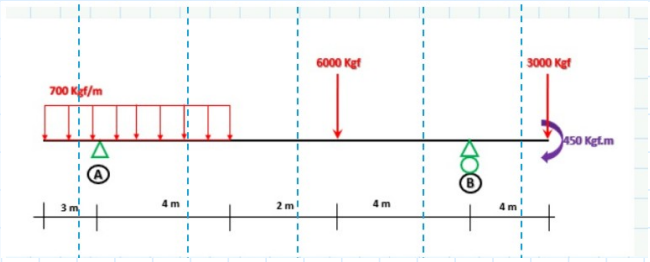
$$B_y := -700 \cdot (7) \cdot (0.5) - 6000 \cdot (6) + B_y \cdot 10 - 3000 \cdot (14) - 450 = 0 \xrightarrow{solve,\, B_y} 8090.0$$

$$A_y := -700 \cdot (7) + A_y - 6000 + (B_y) - 3000 = 0 \xrightarrow{solve,\, A_y} 5810.0$$

$A_y = 5810$

$B_y = 8090$

03. GRAFICAS DE CORTANTE Y MOMENTO

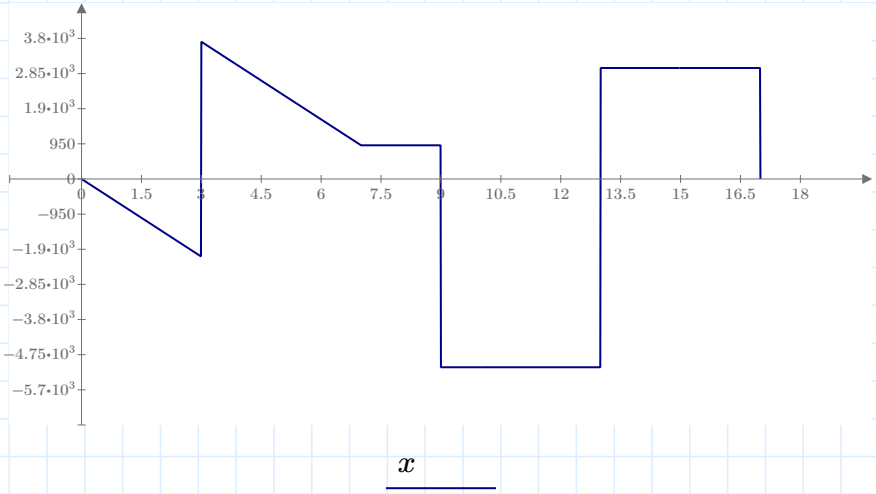


$$V(x) := -700 \cdot x + 700 \cdot (x - 7) \cdot (x \geq 7) + A_y \cdot (x \geq 3) - 6000 \cdot (x \geq 9) + B_y \cdot (x \geq 13) - 3000 \cdot (x \geq 17)$$

$$M(x) := -700 \cdot \left(\frac{x^2}{2}\right) + 700 \cdot \frac{(x-7)^2}{2} \cdot (x \geq 7) + A_y \cdot (x - 3) \cdot (x \geq 3) - 6000 \cdot (x - 9) \cdot (x \geq 9) + B_y \cdot (x - 13) \cdot (x \geq 13) - 3000 \cdot (x - 17) \cdot (x \geq 17) + 450$$

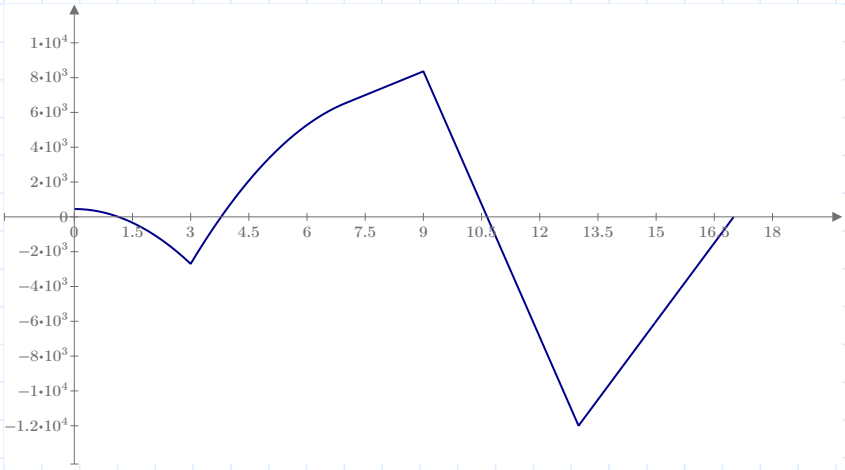
INTERVALOS:

$x := 0, 0.01 .. 17$



$V(x)$

x


1·10^4
8·10^3
6·10^3
4·10^3
2·10^3
0
−2·10^3
−4·10^3
−6·10^3
−8·10^3
−1·10^4
−1.2·10^4
0
1.5
3
4.5
6
7.5
9
10.5
12
13.5
15
16.5
18
M(x)
x